

Ex.no:3	DATA CLEANING

Aim:**Algorithm:**

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows/columns containing any missing value.
7. Use how='all' to remove rows/columns only if all values are missing.
8. Use thresh=N to keep rows/columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
# Unzip the uploaded file
!unzip -o '/content/Diamonds Prices2022.csv.zip'
# Load the dataset into a pandas DataFrame
df = pd.read_csv('/content/Diamonds Prices2022.csv')

print("Dataset loaded successfully. Displaying the first 5 rows:")
display(df.head())

print("\nBasic information about the dataset:")
display(df.info())
# Identify and count missing values
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0]

if not missing_values.empty:
    print("Missing values in each column:")
    display(missing_values)
else:
    print("No missing values found in the dataset.")
    # Count initial number of rows
    initial_rows = df.shape[0]

    # Remove duplicate rows
    df.drop_duplicates(inplace=True)

    # Count rows after removing duplicates
    rows_after_duplicates = df.shape[0]

    # Report the number of duplicates removed
    duplicates_removed = initial_rows - rows_after_duplicates
```

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if duplicates_removed > 0:
    print(f'Successfully removed {duplicates_removed} duplicate rows.')
else:
    print("No duplicate rows found in the dataset.")

print(f'Dataset shape after removing duplicates: {df.shape}')
print("Detecting outliers using the IQR method:")

# Exclude 'Unnamed: 0' as it's typically an index and not a feature for outlier analysis
numerical_cols_for_outliers = [col for col in numerical_cols if col != 'Unnamed: 0']

outlier_summary = {}

for col in numerical_cols_for_outliers:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Identify outliers
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

    if not outliers.empty:
        outlier_summary[col] = {
            'count': len(outliers),
            'percentage': (len(outliers) / len(df)) * 100
        }
    else:
        outlier_summary[col] = {'count': 0, 'percentage': 0.0}

# Display the summary of outliers
if outlier_summary:

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print("\nSummary of Outliers by Column (IQR Method):")

for col, data in outlier_summary.items():

    print(f"- {col}: {data['count']} outliers ({data['percentage']:.2f}% of total data)")

else:

    print("No numerical columns found or no outliers detected by the IQR method.")

```

Output:

```

Dataset loaded successfully. Displaying the first 5 rows:

```

Unnamed: 0	carat	cut	color	clarity	depth	table	price	x	y	z
0	1	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	2.43
1	2	0.21	Premium	E	SI1	59.8	61.0	326	3.89	2.31
2	3	0.23	Good	E	VSI	56.9	65.0	327	4.05	2.31
3	4	0.29	Premium	I	VS2	62.4	58.0	334	4.20	2.63
4	5	0.31	Good	J	SI2	63.3	58.0	335	4.34	2.75

```

Basic information about the dataset:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53943 entries, 0 to 53942
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0  53943 non-null  int64
1   carat       53943 non-null  float64
2   cut         53943 non-null  object
3   color       53943 non-null  object
4   clarity     53943 non-null  object
5   depth       53943 non-null  float64
6   table       53943 non-null  float64
7   price       53943 non-null  int64
8   x           53943 non-null  float64
9   y           53943 non-null  float64
10  z           53943 non-null  float64
dtypes: float64(6), int64(2), object(3)
memory usage: 4.5+ MB
None

```

```

No duplicate rows found in the dataset.
Dataset shape after removing duplicates: (53943, 11)

```

```

Detecting outliers using the IQR method:

Summary of Outliers by Column (IQR Method):
- carat: 1889 outliers (3.50% of total data)
- depth: 2545 outliers (4.72% of total data)
- table: 605 outliers (1.12% of total data)
- price: 3540 outliers (6.56% of total data)
- x: 32 outliers (0.06% of total data)
- y: 29 outliers (0.05% of total data)
- z: 49 outliers (0.09% of total data)

```

Result: